

Cross-Sectional Price Momentum Long-Short Strategy

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ECO 43000-ONL | Prof. John Droescher | City College of New York | May 26, 2026

INTRODUCTION:

The purpose of this report is to evaluate whether a systematic momentum-based trading strategy can generate risk-adjusted excess returns, or alpha, relative to a passive benchmark using only publicly available price and volume data. More broadly, the study explores key ideas in quantitative portfolio management, including persistent market anomalies, dollar-neutral long-short portfolio construction, and factor-based ranking systems for stock selection.

We implement a monthly rebalanced long-short equity strategy applied to the 101 constituents of the Nasdaq-100, represented by the Invesco QQQ ETF. The strategy uses a five-factor composite momentum signal based on Berry Cox-style price momentum indicators. At each month-end, it ranks stocks by their composite momentum scores, goes long the highest-ranked stocks, and shorts the lowest-ranked stocks.

Long-short equity strategies are designed to generate returns from relative performance between assets rather than from the overall direction of the market. This project focuses on cross-sectional price momentum: the tendency for recent outperformers to continue outperforming and recent underperformers to continue lagging. The strategy is tested over a trailing five-year backtest period from 2021 to 2026 and benchmarked against QQQ.

METHODOLOGY:

Asset Universe Construction:

The asset universe construction was defined using Nasdaq-100 constituents from `qqq_holdings.csv`.

Historical Data Collection:

Using `yfinance`, we collected 10 years of daily price and volume data.

Factor Engineering:

- 52-Week Trend Slope
- Percent Above 260-Day Low
- Price Oscillator
- 39-Week Return
- 51-Week Volume Price Trend (VPT)

Cross-Sectional Signal Construction:

All factors were standardized using cross-sectional z-scores at each month-end and averaged into a composite score.

Portfolio Construction:

- Long: Top 20 momentum stocks
- Short: Bottom 5 momentum stocks
- Conviction weighting based on signal strength

Backtest Design and Look-Ahead Prevention :

The backtest simulates monthly rebalancing over five years while avoiding look-ahead bias through a one-period signal shift. The composite score calculated at the end of month **M** is used to form the portfolio for the following month, with returns earned from the end of month **M** to the end of month **M+1**. This is implemented in pandas with `signals.shift(1)`, ensuring the strategy only uses information available at the time of trading. The 20-day factor lags provide an additional layer of protection against future data leakage.

RESULTS:

	Portfolio	Long	Short	Benchmark
CAGR	0.383587	0.541340	0.197740	0.166490
AnnVol	0.322955	0.402879	0.316875	0.205267
Sharpe	1.158761	1.268703	0.726544	0.855783
MaxDD	-0.264000	-0.332545	-0.506825	-0.325770

Table 1. Strategic Performance Metrics Summary

Table 1 summarizes the backtest performance of the Long/Short portfolio compared to its Long-only, Short-only, and Benchmark (QQQ) allocations. The strategy delivers strong excess returns, with a CAGR of 38.36% versus 16.65% for the benchmark. It also shows improved risk-adjusted performance, achieving a Sharpe Ratio of 1.16 compared to 0.86 for QQQ. Volatility remains higher than the benchmark but is controlled through diversification across the long and short baskets. Most importantly, downside risk is reduced, with a maximum drawdown of -26.40%, outperforming the benchmark and demonstrating the effectiveness of the hedging structure.

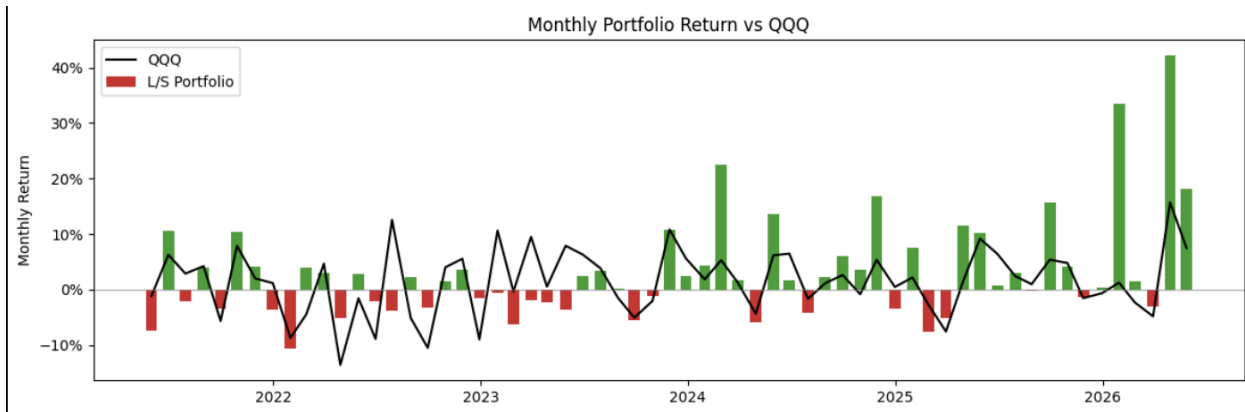


Figure 1. Monthly Portfolio Return vs QQQ

Figure 1 shows the monthly return distribution of the strategy. Most of the period reflects positive excess returns relative to the benchmark, indicating consistent alpha generation. The green bars during market drawdowns highlight the hedge working in practice, helping cushion downside risk. Overall, it shows the system is not just capturing upside momentum but also staying relatively stable during volatile or negative market conditions.

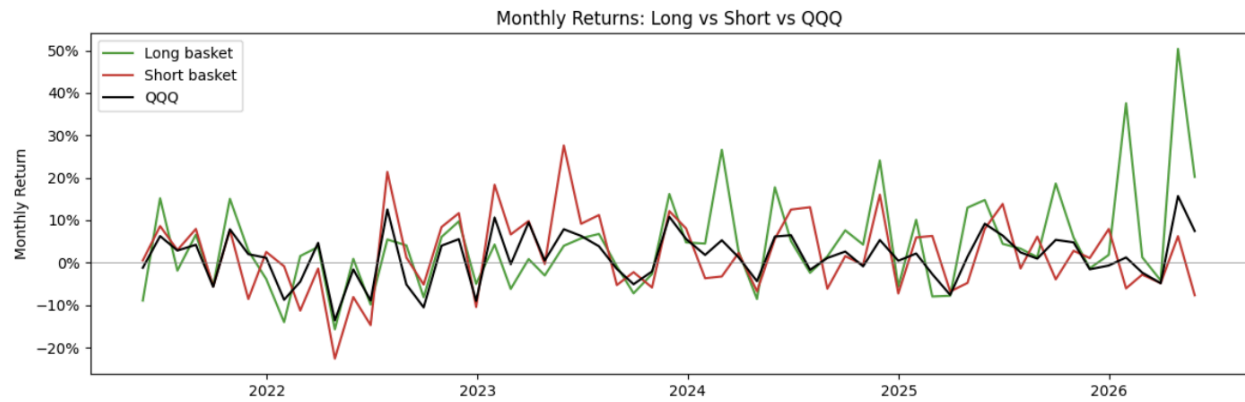


Figure 2. Monthly Returns: Long vs Short vs QQQ

Figure 2 shows the monthly return paths for the Long basket, Short basket, and QQQ. This helps visualize how each leg of the strategy behaves relative to the benchmark over time. The Long basket generally tracks above QQQ during upward market moves, showing that the momentum signals are capturing strong performers. The Short basket moves in the opposite direction and tends to diverge from QQQ in weaker periods, which supports its role in generating returns when markets decline and helping balance overall portfolio exposure.

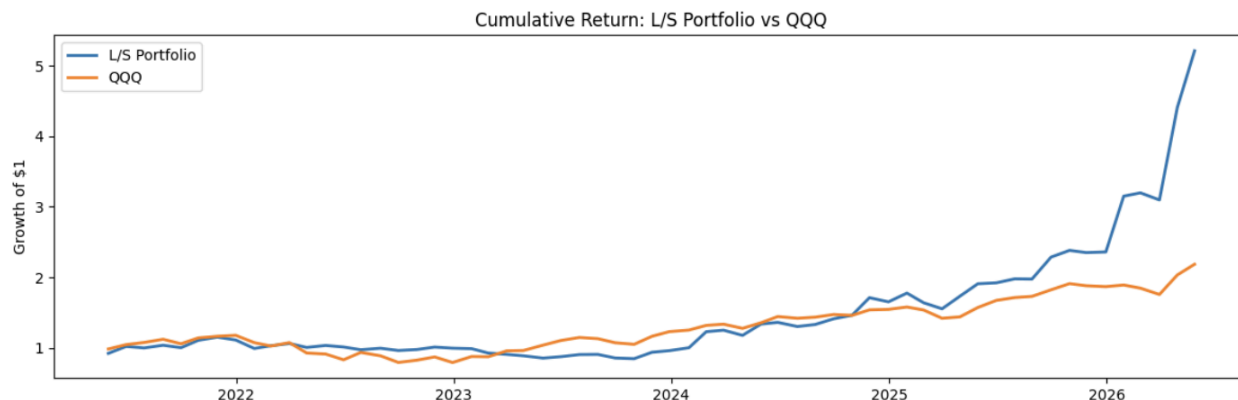


Figure 3. Cumulative Return: L/S Portfolio vs QQQ

Figure 3 displays the cumulative growth of the Long/Short portfolio compared to QQQ over the five-year backtest period. Starting from a hypothetical \$1 investment, the chart shows the portfolio consistently compounding at a faster rate than the benchmark. The growing gap between the two curves highlights the effectiveness of the monthly momentum-based restructuring process and demonstrates the strategy's ability to generate stronger long-term returns over time.

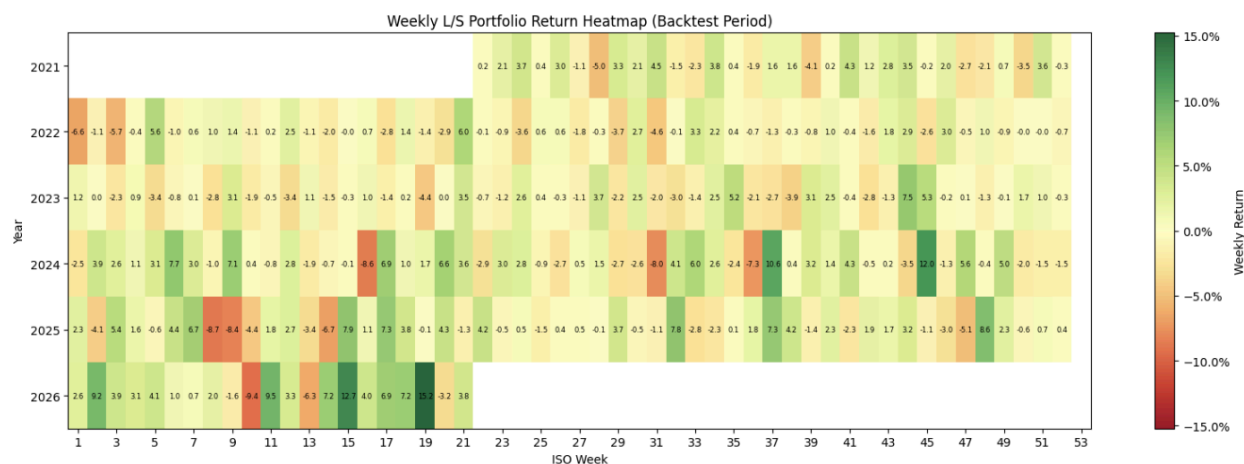


Figure 4. Weekly L/S Portfolio Return Heatmap

Figure 4 shows the weekly return heatmap of the Long/Short portfolio across different years and ISO weeks. The heatmap makes it easier to spot periods of strong performance, weaker stretches, and overall consistency throughout the backtest. Clusters of green weeks reflect sustained positive momentum, while the limited red periods show the strategy's ability to recover relatively quickly from short-term drawdowns and changing market conditions.

DISCUSSION:

The strategy significantly outperformed the benchmark in both return and risk-adjusted terms. The inclusion of a short book improved downside protection and reduced drawdowns relative to the long-only exposure.

LIMITATIONS:

- **Transaction costs and slippage are not modeled.** Monthly rebalancing across 25 positions would likely incur commissions, bid-ask spreads, and execution costs, all of which would reduce realized alpha. However, the magnitude of the strategy's excess returns may provide some buffer.
- **Short borrowing costs are excluded.** Stocks selected for the short leg may carry elevated borrowing fees, particularly if they are already heavily shorted. These costs would reduce the net contribution of the short portfolio in practice.
- **Survivorship bias may be present.** The investment universe is based on the current QQQ constituent list, excluding companies that were removed from the Nasdaq-100 during the backtest period. This may modestly overstate performance, as removed constituents may have disproportionately included weaker momentum stocks.
- **Momentum crash risk remains a limitation.** Sharp market reversals or sudden shifts in investor sentiment can produce significant losses, particularly in the short leg. Such episodes highlight the vulnerability of pure momentum strategies during rapid factor reversals.

CONCLUSION:

This project shows that a systematic cross-sectional momentum strategy based on Berry Cox's five price momentum factors can generate strong risk-adjusted alpha within the Nasdaq-100 universe. Over the five-year backtest, the strategy produced a 38.36% CAGR, a Sharpe ratio of 1.16, and a maximum drawdown of -26.40%, outperforming QQQ on a risk-adjusted basis. The results support both the persistence of momentum and the effectiveness of the strategy's design choices, including cross-sectional z-score normalization, 20-day factor lags, conviction-weighted sizing, and a one-period signal shift to prevent look-ahead bias. Most importantly, the project highlights that a backtest is only useful if it reflects what a real trader could have known and acted on at the time. Future improvements could include transaction costs, volatility targeting, and testing across a broader stock universe.

References

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